PRINCIPLES OF DATA SCIENCE

PRACTICAL RECORD

##241801282

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Course Code: AD23532

Semester: V Semester – B.Tech AIDS

Department: Artificial Intelligence and Data Science

EXPERIMENT 1 – DATA PREPROCESSING AND CLEANING

AIM

To clean the Telco Customer Churn dataset, handle missing values and duplicates, encode categorical data, standardise numeric features, and plot important relationships.

ALGORITHM

1. Import the required libraries and load the Telco CSV.
2. Inspect first/last rows, shape, columns, data types, summary statistics, missing values and duplicates.

4. Standardise Tenure, MonthlyCharges and TotalCharges; encode categorical variables.
5. Plot distributions, Contract versus Churn, MonthlyCharges by Churn and numeric correlation heatmap.

In [5]

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```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.preprocessing import StandardScaler

df = pd.read_csv(class="s">"Telco-Customer-Churn.csv.csv")
print(df.head(10), df.tail(10), df.shape, df.columns.tolist())
print(df.info(), df.describe(include=class="s">"all"))
print(df.isnull().sum(), class="s">"Duplicates:", df.duplicated().sum())
df = df.drop_duplicates().copy()
df[class="s">"TotalCharges"] = pd.to_numeric(df[class="s">"TotalCharges"], erro
for c in df.select_dtypes(include=class="s">"number"):
    df[c] = df[c].fillna(df[c].median())
for c in df.select_dtypes(exclude=class="s">"number"):
    if df[c].isnull().any(): df[c] = df[c].fillna(df[c].mode()[0])
for c in [class="s">"tenure", class="s">"MonthlyCharges", class="s">"TotalCharg
    df[c] = StandardScaler().fit_transform(df[[c]])
sns.histplot(df[class="s">"MonthlyCharges"], kde=True); plt.show()
sns.histplot(df[class="s">"tenure"], kde=True); plt.show()
sns.countplot(data=df, x=class="s">"Contract", hue=class="s">"Churn"); plt.xtic
sns.boxplot(data=df, x=class="s">"Churn", y=class="s">"MonthlyCharges"); plt.sh
sns.heatmap(df.select_dtypes(class="s">"number").corr(), annot=True); plt.show(
df = pd.get_dummies(df, drop_first=True, dtype=int)
```

```
   customerID  gender  SeniorCitizen Partner Dependents  tenure PhoneService
0  7590-VHVEG  Female              0     Yes         No       1           No
1  5575-GNVDE    Male              0      No         No      34          Yes
2  3668-QPYBK    Male              0      No         No       2          Yes
3  7795-CFOCW    Male              0      No         No      45           No
```

```
7  6713-OKOMC  Female              0      No         No      10           No
8  7892-POOKP  Female              0     Yes         No      28          Yes
9  6388-TABGU    Male              0      No        Yes      62          Yes

      MultipleLines InternetService OnlineSecurity  ... DeviceProtection  \
0  No phone service             DSL             No  ...               No
1                No             DSL            Yes  ...              Yes
2                No             DSL            Yes  ...               No
3  No phone service             DSL            Yes  ...              Yes
4                No     Fiber optic             No  ...               No
5               Yes     Fiber optic             No  ...              Yes
6               Yes     Fiber optic             No  ...               No
7  No phone service             DSL            Yes  ...               No
8               Yes     Fiber optic             No  ...              Yes
9                No             DSL            Yes  ...               No

  TechSupport StreamingTV StreamingMovies        Contract PaperlessBilling
0          No          No              No  Month-to-month              Yes
1          No          No              No        One year               No
2          No          No              No  Month-to-month              Yes
3         Yes          No              No        One year               No
4          No          No              No  Month-to-month              Yes
5          No         Yes             Yes  Month-to-month              Yes
6          No         Yes              No  Month-to-month              Yes
7          No          No              No  Month-to-month               No
8         Yes         Yes             Yes  Month-to-month              Yes
9          No          No              No        One year               No

               PaymentMethod MonthlyCharges  TotalCharges Churn
0           Electronic check          29.85         29.85    No
1               Mailed check          56.95        1889.5    No
2               Mailed check          53.85        108.15   Yes
3  Bank transfer (automatic)          42.30       1840.75    No
4           Electronic check          70.70        151.65   Yes
5           Electronic check          99.65         820.5   Yes
6    Credit card (automatic)          89.10        1949.4    No
7               Mailed check          29.75         301.9    No
8           Electronic check         104.80       3046.05   Yes
9  Bank transfer (automatic)          56.15       3487.95    No
```

```
      7034  0639-TSIQW  Female              0      No         No      67
      7035  8456-QDAVC    Male              0      No         No      19
      7036  7750-EYXWZ  Female              0      No         No      12
      7037  2569-WGERO  Female              0      No         No      72
      7038  6840-RESVB    Male              0     Yes        Yes      24
      7039  2234-XADUH  Female              0     Yes        Yes      72
      7040  4801-JZAZL  Female              0     Yes        Yes      11
      7041  8361-LTMKD    Male              1     Yes         No       4
      7042  3186-AJIEK    Male              0      No         No      66

           PhoneService     MultipleLines InternetService       OnlineSecurity  ..
      7033          Yes                No     Fiber optic                   No  ..
      7034          Yes               Yes     Fiber optic                  Yes  ..
      7035          Yes                No     Fiber optic                   No  ..
      7036           No  No phone service             DSL                   No  ..
      7037          Yes                No              No  No internet service  ..
      7038          Yes               Yes             DSL                  Yes  ..
      7039          Yes               Yes     Fiber optic                   No  ..
      7040           No  No phone service             DSL                  Yes  ..
      7041          Yes               Yes     Fiber optic                   No  ..
      7042          Yes                No     Fiber optic                  Yes  ..

              DeviceProtection          TechSupport          StreamingTV  \
      7033                  No                   No                  No
      7034                 Yes                   No                 Yes
      7035                  No                   No                 Yes
      7036                 Yes                  Yes                 Yes
      7037  No internet service  No internet service  No internet service
      7038                 Yes                  Yes                 Yes
      7039                 Yes                   No                 Yes
      7040                  No                   No                  No
      7041                  No                   No                  No
      7042                 Yes                  Yes                 Yes

               StreamingMovies        Contract PaperlessBilling  \
      7033                  No  Month-to-month              Yes
      7034                  No  Month-to-month              Yes
      7035                  No  Month-to-month              Yes
      7036                 Yes        One year               No
```

```
7040                   No  Month-to-month              Yes
7041                   No  Month-to-month              Yes
7042                  Yes        Two year              Yes

                  PaymentMethod MonthlyCharges  TotalCharges Churn
7033    Credit card (automatic)          69.50       2625.25    No
7034    Credit card (automatic)         102.95       6886.25   Yes
7035  Bank transfer (automatic)          78.70        1495.1    No
7036           Electronic check          60.65         743.3    No
7037  Bank transfer (automatic)          21.15        1419.4    No
7038               Mailed check          84.80        1990.5    No
7039    Credit card (automatic)         103.20        7362.9    No
7040           Electronic check          29.60        346.45    No
7041               Mailed check          74.40         306.6   Yes
7042  Bank transfer (automatic)         105.65        6844.5    No

[10 rows x 21 columns] (7043, 21) ['customerID', 'gender', 'SeniorCitizen',
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 7043 entries, 0 to 7042
Data columns (total 21 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   customerID        7043 non-null   object
 1   gender            7043 non-null   object
 2   SeniorCitizen     7043 non-null   int64
 3   Partner           7043 non-null   object
 4   Dependents        7043 non-null   object
 5   tenure            7043 non-null   int64
 6   PhoneService      7043 non-null   object
 7   MultipleLines     7043 non-null   object
 8   InternetService   7043 non-null   object
 9   OnlineSecurity    7043 non-null   object
 10  OnlineBackup      7043 non-null   object
 11  DeviceProtection  7043 non-null   object
 12  TechSupport       7043 non-null   object
 13  StreamingTV       7043 non-null   object
 14  StreamingMovies   7043 non-null   object
 15  Contract          7043 non-null   object
 16  PaperlessBilling  7043 non-null   object
```

```
 20  Churn             7043 non-null   object
dtypes: float64(1), int64(2), object(18)
memory usage: 1.1+ MB
None         customerID gender  SeniorCitizen Partner Dependents       tenur
count          7043   7043    7043.000000    7043       7043  7043.000000
unique         7043      2            NaN       2          2          NaN
top     3186-AJIEK   Male            NaN      No         No          NaN
freq              1   3555            NaN    3641       4933          NaN
mean            NaN    NaN       0.162147     NaN        NaN    32.371149
std             NaN    NaN       0.368612     NaN        NaN    24.559481
min             NaN    NaN       0.000000     NaN        NaN     0.000000
25%             NaN    NaN       0.000000     NaN        NaN     9.000000
50%             NaN    NaN       0.000000     NaN        NaN    29.000000
75%             NaN    NaN       0.000000     NaN        NaN    55.000000
max             NaN    NaN       1.000000     NaN        NaN    72.000000

       PhoneService MultipleLines InternetService OnlineSecurity  ...  \
count          7043          7043            7043           7043  ...
unique            2             3               3              3  ...
top             Yes            No     Fiber optic             No  ...
freq           6361          3390            3096           3498  ...
mean            NaN           NaN             NaN            NaN  ...
std             NaN           NaN             NaN            NaN  ...
min             NaN           NaN             NaN            NaN  ...
25%             NaN           NaN             NaN            NaN  ...
50%             NaN           NaN             NaN            NaN  ...
75%             NaN           NaN             NaN            NaN  ...
max             NaN           NaN             NaN            NaN  ...

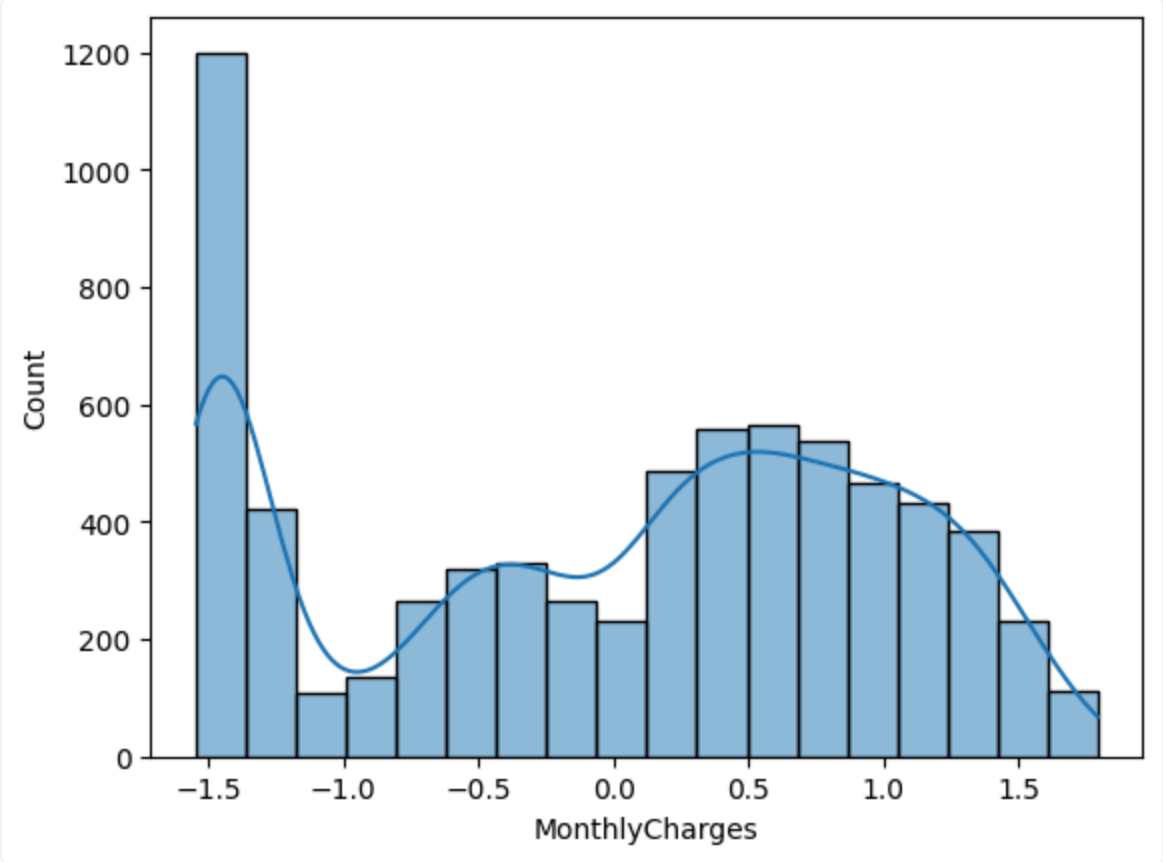
       DeviceProtection TechSupport StreamingTV StreamingMovies  \
count              7043        7043        7043            7043
unique                3           3           3               3
top                  No          No          No              No
freq               3095        3473        2810            2785
mean                NaN         NaN         NaN             NaN
std                 NaN         NaN         NaN             NaN
min                 NaN         NaN         NaN             NaN
25%                 NaN         NaN         NaN             NaN
50%                 NaN         NaN         NaN             NaN
```

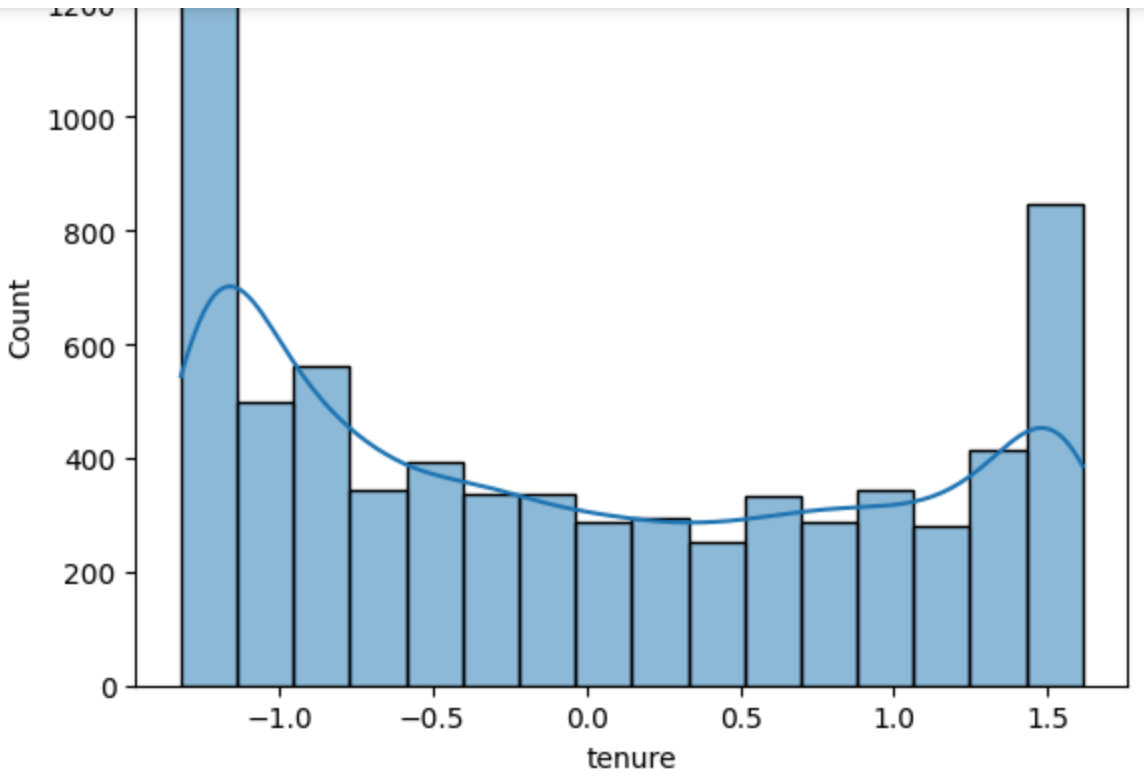
```
              Contract PaperlessBilling     PaymentMethod MonthlyCharges  \
count             7043             7043              7043    7043.000000
unique               3                2                 4            NaN
top     Month-to-month              Yes  Electronic check            NaN
freq              3875             4171              2365            NaN
mean               NaN              NaN               NaN      64.761692
std                NaN              NaN               NaN      30.090047
min                NaN              NaN               NaN      18.250000
25%                NaN              NaN               NaN      35.500000
50%                NaN              NaN               NaN      70.350000
75%                NaN              NaN               NaN      89.850000
max                NaN              NaN               NaN     118.750000

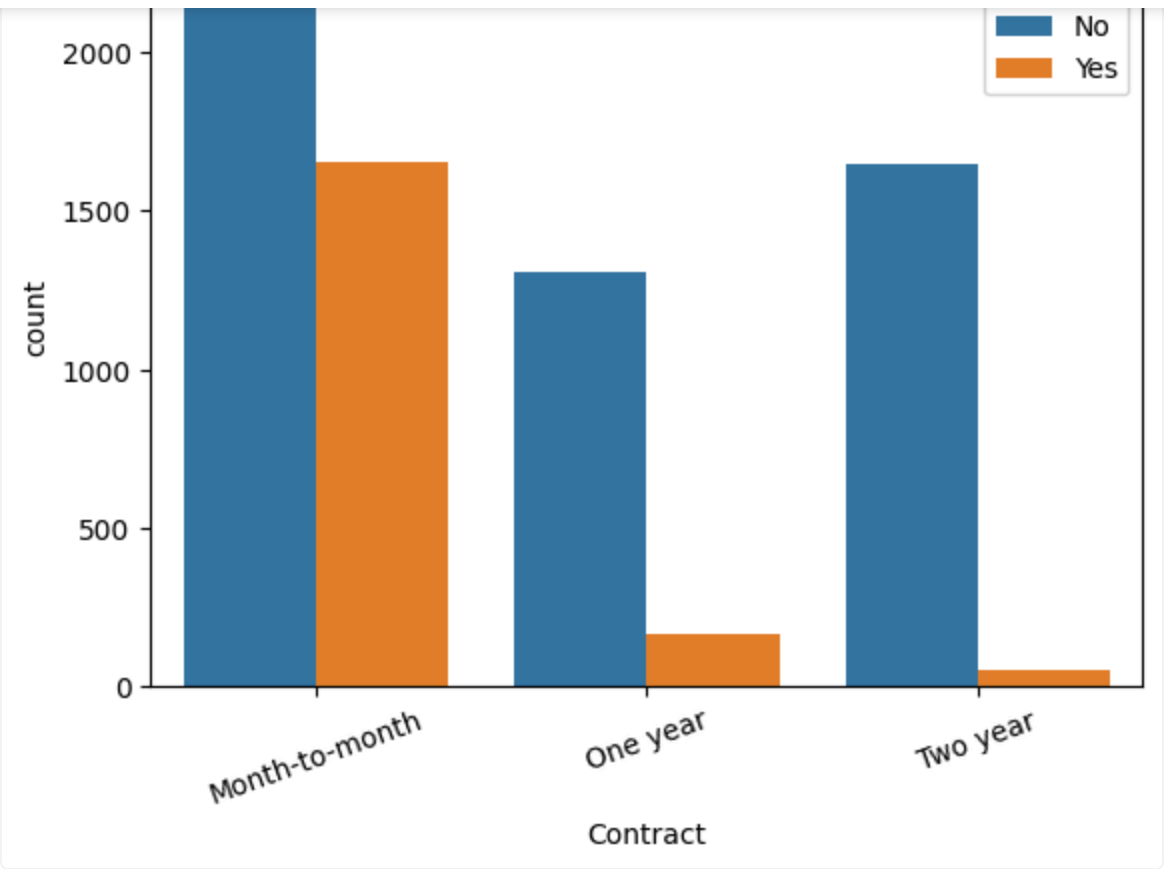
        TotalCharges Churn
count           7043  7043
unique          6531     2
top                    No
freq              11  5174
mean             NaN   NaN
std              NaN   NaN
min              NaN   NaN
25%              NaN   NaN
50%              NaN   NaN
75%              NaN   NaN
max              NaN   NaN

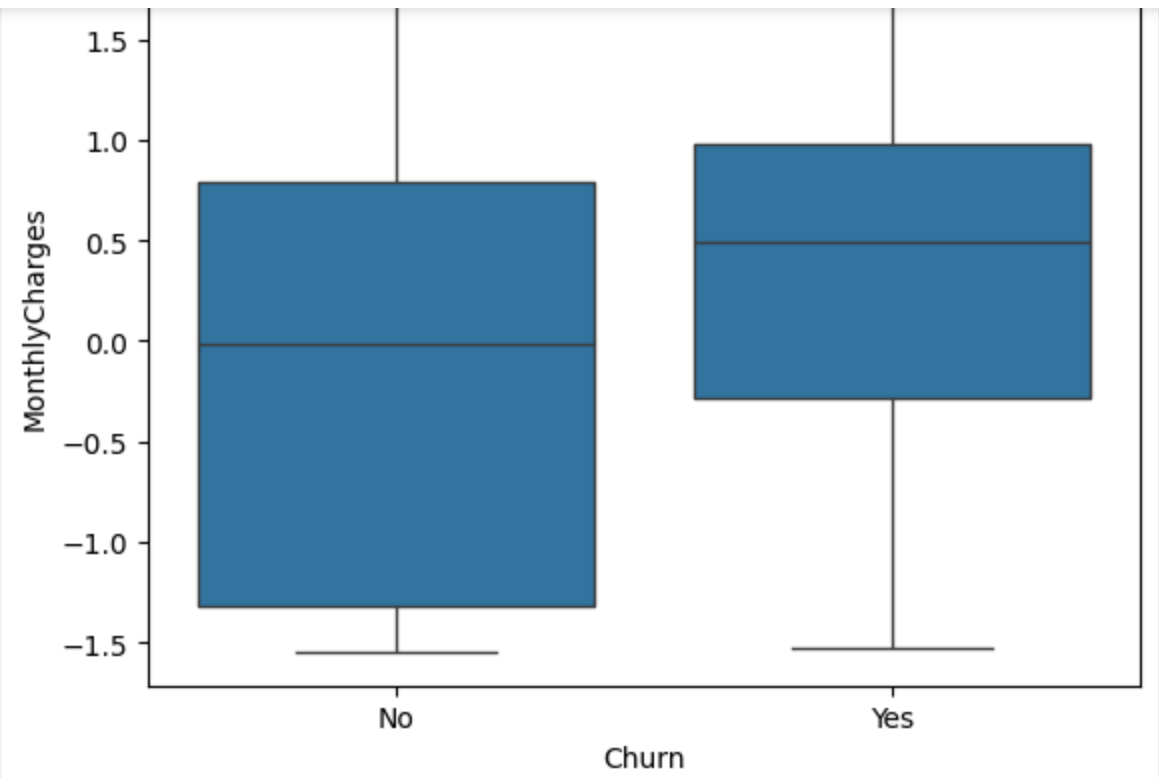
[11 rows x 21 columns]
customerID          0
gender              0
SeniorCitizen       0
Partner             0
Dependents          0
tenure              0
PhoneService        0
MultipleLines       0
InternetService     0
OnlineSecurity      0
OnlineBackup        0
DeviceProtection    0
```

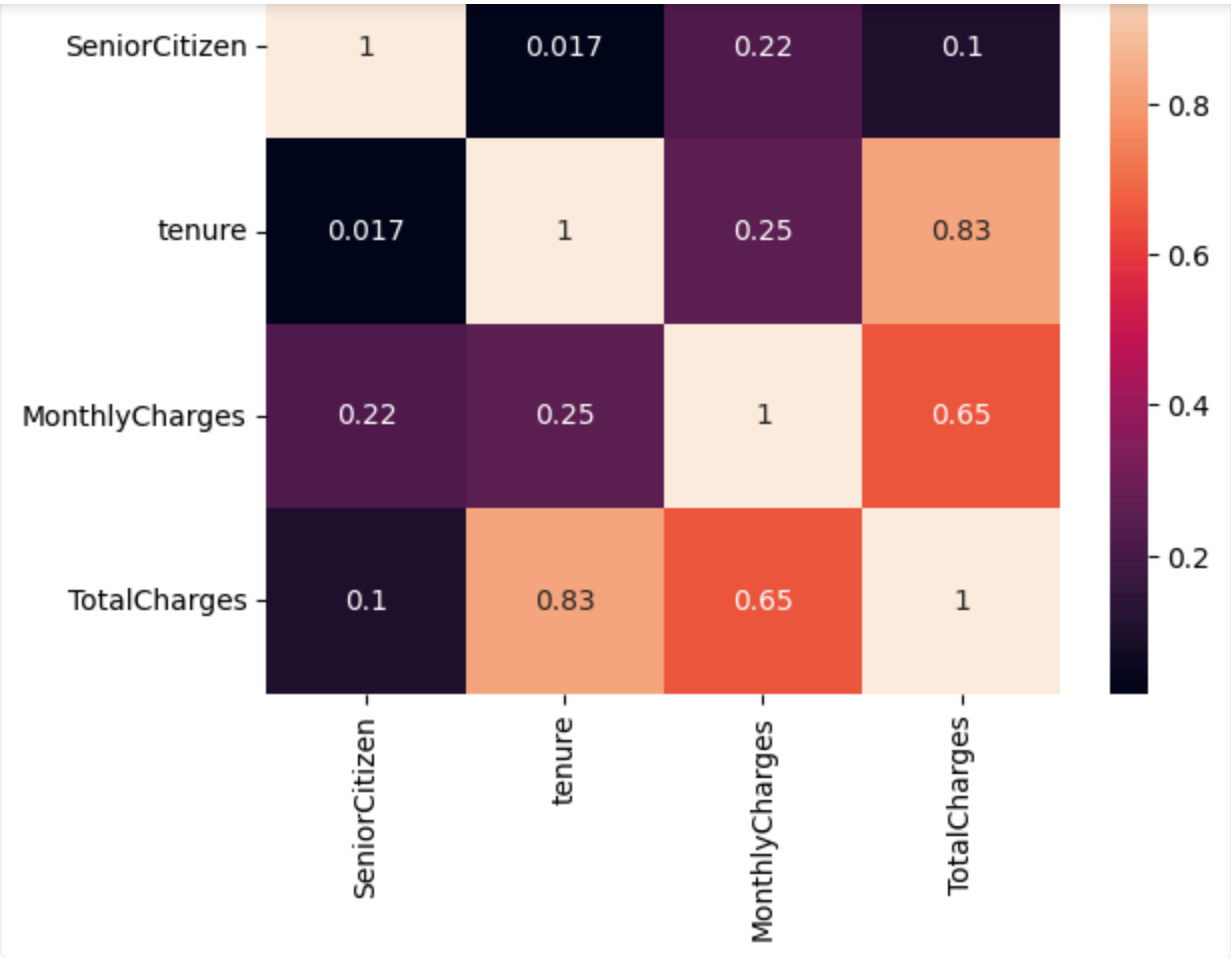
```
Contract            0
PaperlessBilling    0
PaymentMethod       0
MonthlyCharges      0
TotalCharges        0
Churn               0
dtype: int64 Duplicates: 0
```




1000
800
600
400
200
0
Count
−1.0
−0.5
0.0
0.5
1.0
1.5
tenure


No
Yes
2000
1500
1000
500
0
count
Month-to-month
One year
Two year
Contract


1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
MonthlyCharges
No
Yes
Churn



RESULT

Thus, the dataset is cleaned and prepared, numerical features are standardised, categorical variables are encoded, and the required plots are generated.